

ACC250: Intro to Financial Accounting
Ch3. The Income Statement

Jaeyoon Yu, Ph.D.
Central Michigan University

- Extend the debit/credit framework to revenues and expenses.
- Distinguish accrual-basis from cash-basis accounting.
- Apply the revenue recognition principle and the matching principle.
- Account for receivables, payables, deferred revenues, and prepaid expenses.

1 Introduction to Income Statement and Accrual Accounting

- Operating Activities and Cycle
- Revenues, Expenses, and Net Income
- Debit-Credit Framework - Expansion
- Accrual-basis Accounting vs Cash-basis Accounting

2 Important Accounts

- Accounts Receivable (Assets)
- Accounts Payable (Liabilities)
- Deferred Revenues (Unearned Revenues, Liabilities)
- Prepaid Expenses (Assets)

3 Revenue and Expense Recognition - Cases

4 Illustration

- Journal Entries
- T-accounts
- Unadjusted trial balance

5 Other topics

- I/S - temporary & B/S - permanent
- Net Profit Margin

Opening illustration

1. Your company has sold a product to a customer on cash in December. The product was delivered and cash collected in December.

- Revenues: December.
- Cash inflows: December.

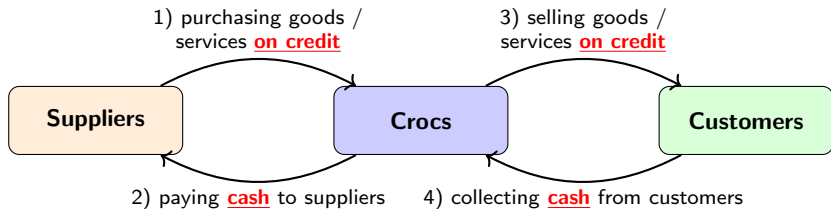
2. Your company has sold a product to a customer on cash in December. The product was delivered in December but cash collected in January.

- Revenues: December.
- Cash inflows: January.

In this chapter, we study when to recognize Revenues and Expenses.

Operating activities:

- the day-to-day activities of a business.
- primary sources of Revenues and Expenses.
- mainly summarized in the Income Statement.
- include:



Operating cycle:

- the period from the purchase of goods and services to the collection of cash from customers.

Revenues

Revenues (R) - Simple

The amount of resources (cash) earned from selling goods or services for a period.
(“earned” means products & services are delivered to customers.)

Revenues (R) - Formal

Increases in assets or decreases in liabilities that arise when a company provides goods or services.

Revenue Recognition Principle

Revenue is recognized when products & services are delivered, regardless of when cash is received.

The formal definition may not make sense to you. We will visit this later in this packet.¹

¹The textbook presents revenue recognition based on the five-step model under the new standard (ASC 606). We do not cover the five-step model here; the principle stated above is enough at the intro level.

How much revenue should Noodlecake recognize in September?

- ① In September, Noodlecake sold \$3K of apps online. Cash collected in September.
- ② Noodlecake issued three \$100 gift cards at the beginning of September. These gift cards have not been used at the end of September.
- ③ Noodlecake sold \$9K of apps in the App Store and on Google Play in September. Cash will be collected in October.

- Revenues: \$12K (3K + 0 + 9K)
- Cash flow timing isn't important for revenue recognition.

Expenses

Expenses (E) - Simple

Costs of operating a business, incurred to generate revenues in the period.

Expenses (E) - Formal

Decreases in assets or increases in liabilities relating to costs a company incurs when providing goods or services.

Matching Principle

- Expenses should be recognized in the same period as the revenues they help to generate, regardless of when cash is paid. In other words,
- Expenses should be recognized when the resources are used up to generate the revenues.

The formal definition may not make sense to you. We will visit this later in this packet.

- ① When a company buys supplies, it's assets (A).
- ② When the supplies are used up to generate revenues, expense should be recognized expenses (E).

EX. When to recognize expenses?

- ① Noodlecake bought \$1K of supplies in September.
- ② Noodlecake used up the supplies in October.

Answer: October.

How much expenses should Noodlecake recognize in September?

- 1 Noodlecake wrote checks to employees totaling \$7,800 for wages related to hours worked in September.
- 2 Noodlecake displayed online ads in Sep and received a bill for \$500 to be paid in October.
- 3 Noodlecake was notified by its bank that an automatic monthly payment of \$600 was transmitted to its utility company for electricity use in September.

- Expenses: $\$7,800 + \$500 + \$600 = \$8,900$
- Cash flow timing isn't important for expense recognition.
- Expenses should be recognized in the period when the goods or services are used up.

Net Income (NI)

Revenues - Expenses.

- Net Income is a measure of a company's profitability.
- Net Income is NOT an account.
- Net Income is NOT the same as cash flows from operating activities.

$$A_t = L_t + SHE_t$$

$$A_t = L_t + (CC_t + RE_t)$$

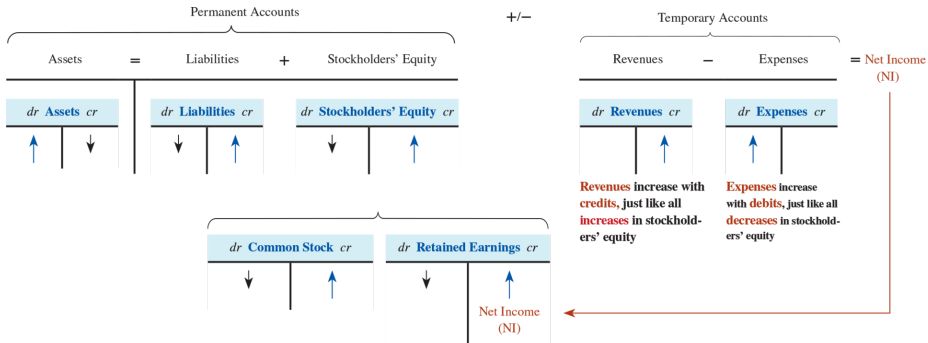
$$A_t = L_t + CC_t + (RE_{t-1} + NI_t - D_t)$$

$$A_t = L_t + CC_t + (RE_{t-1} + R_t - E_t - D_t)$$

$$A_t + E_t + D_t = L_t + CC_t + (RE_{t-1} + R_t)$$

<i>A</i>	Assets
<i>L</i>	Liabilities
<i>SHE</i>	Stockholders' Equity
<i>CC</i>	Contributed Capital
<i>RE</i>	Retained Earnings
<i>R</i>	Revenues
<i>E</i>	Expenses
<i>NI</i>	Net Income
<i>D</i>	Dividends

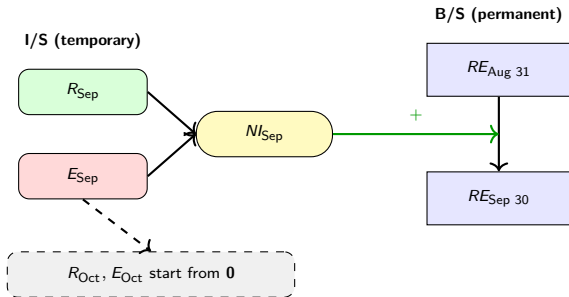
Debit-Credit Framework - Expansion



- RE: **credit balance** (Ch2).
- R: **credit balance** as R has a positive relationship with RE.
- E: **debit balance** as E has a negative relationship with RE.

EX. Does it make sense to recognize September's R & E once again in October's R & E?

- **No.** September's R & E belong to September only.
- At the end of September, $R - E = NI$, which is absorbed into Retained Earnings.
- R & E are then cleared to 0, ready to take new records from October transactions.



Accrual-basis Accounting vs Cash-basis Accounting

- Accrual-basis accounting recognizes R & E when they are earned or incurred, regardless of when cash is received or paid.
- Cash-basis accounting recognizes R & E when cash is received or paid.
- Most companies rely on credit-based transactions, so accrual-basis accounting is more accurate.

Accrual-basis Accounting vs Cash-basis Accounting

- Noodlecake paid its employees \$10K in January.
- \$15K credit sales were made in January, but it was collected in February.

Cash-basis Accounting

	Jan	Feb
Revenues	<u>\$0</u>	<u>\$15,000</u>
Expenses	<u>\$10,000</u>	<u>\$0</u>
Net Income	<u><u>-\$10,000</u></u>	<u><u>\$15,000</u></u>

Accrual-basis Accounting

	Jan	Feb
Revenues	<u>\$15,000</u>	<u>\$0</u>
Expenses	<u>\$10,000</u>	<u>\$0</u>
Net Income	<u><u>\$5,000</u></u>	<u><u>\$0</u></u>

Accrual-basis accounting helps to match R & E in the same period.

- 1 Introduction to Income Statement and Accrual Accounting
 - Operating Activities and Cycle
 - Revenues, Expenses, and Net Income
 - Debit-Credit Framework - Expansion
 - Accrual-basis Accounting vs Cash-basis Accounting
- 2 Important Accounts
 - Accounts Receivable (Assets)
 - Accounts Payable (Liabilities)
 - Deferred Revenues (Unearned Revenues, Liabilities)
 - Prepaid Expenses (Assets)
- 3 Revenue and Expense Recognition - Cases
- 4 Illustration
 - Journal Entries
 - T-accounts
 - Unadjusted trial balance
- 5 Other topics
 - I/S - temporary & B/S - permanent
 - Net Profit Margin

Accounts Receivable (Assets)

When customers buy products from you on credit:

- Products have been delivered, but customers haven't paid yet.
- You have the right to collect the money.
- You record the amount as Accounts Receivable (Asset!).
- The customer firm records the amount as Accounts Payable.
- Example: Credit Sales.

Accounts Receivable (A/R, Asset)

Amounts customers owe you for products and services sold on credit.

Date 1 - Sold on credit

Acc.Rec (+A)	10K
Revenues (+R, +SE)	10K

Date 2 - Collected cash

Cash (+A)	10K
Acc.Rec (-A)	10K

Accounts Payable (Liabilities)

When you buy supplies on credit:

- supplies have been delivered to you, but you haven't paid yet.
- You have the obligation to pay the money.
- You record the amount as Accounts Payable (Liab!).
- The supplier firm records the amount as Accounts Receivable.
- Example: Credit Purchases.

Accounts Payable (Liability)

Amounts owed to suppliers for products and services purchased on credit.

Date 1 - Purchased on credit

Supplies (+A)	10K
Acc.Pay (+L)	10K

Date 2 - Paid cash

Acc.Pay (-L)	10K
Cash (-A)	10K

Deferred Revenues (Unearned Revenues, Liabilities)

When you receive cash in advance for future products and services:

- you've received the money, but you haven't delivered the products and services yet.
- You have the obligation to deliver the products and services.
- You record the amount as Deferred Revenues or Unearned Revenues (Liab!).
- This liability is settled when the products and services are delivered.
- Example: Airline tickets purchased months ago.

Deferred (Unearned) Revenues (Liability)

Cash received before the service is performed or the product is delivered.

Date 1 - Received cash

Cash (+A)	10K
Def.Rev (+L)	10K

Date 2 - Delivered products & services

Def.Rev (-L)	10K
Sales Revenues (+R, +SE)	10K

Prepaid Expenses (Assets)

When you pay for products and services in advance:

- You've paid the money, but you haven't received the products and services yet.
- You have the right to receive the products and services.
- You record the amount as **Prepaid Expenses (Asset!)**.
- This asset is converted to **Expenses** when the products and services are **used up**.
- Example: Prepaid 3-year insurance policy.

Prepaid Expenses (Asset)

Amounts paid in advance for products and services that will be received in the future.

Date 1 - Paid cash for 12 month insurance policy

Prepaid Expense (+A)	12K
Cash (-A)	12K

Date 2 - At the end of month 1: Recognize 1 month of expense (as used up)

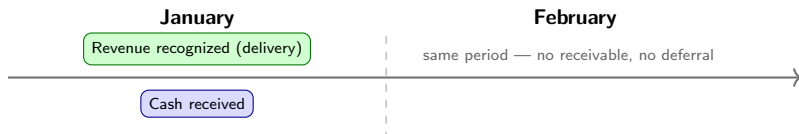
Expenses (+E)	1K
Prepaid Expense (-A)	1K

- 1 Introduction to Income Statement and Accrual Accounting
 - Operating Activities and Cycle
 - Revenues, Expenses, and Net Income
 - Debit-Credit Framework - Expansion
 - Accrual-basis Accounting vs Cash-basis Accounting
- 2 Important Accounts
 - Accounts Receivable (Assets)
 - Accounts Payable (Liabilities)
 - Deferred Revenues (Unearned Revenues, Liabilities)
 - Prepaid Expenses (Assets)
- 3 Revenue and Expense Recognition - Cases
- 4 Illustration
 - Journal Entries
 - T-accounts
 - Unadjusted trial balance
- 5 Other topics
 - I/S - temporary & B/S - permanent
 - Net Profit Margin

Revenue Recognition - Case A

Prepare the **journal entries required at the end of each month:**

ABC corp. sold smartphones and collected \$10K in January.



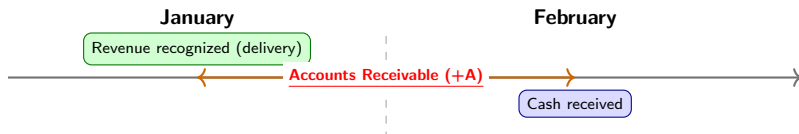
January:

Cash (+A)	10K
Revenues (+R, +SE)	10K

Revenue Recognition - Case B

Prepare the **journal entries required at the end of each month:**

ABC corp. sold smartphones in January and collected \$10K in February.



January:

Acc.Rec (+A)	10K
Revenues (+R, +SE)	10K

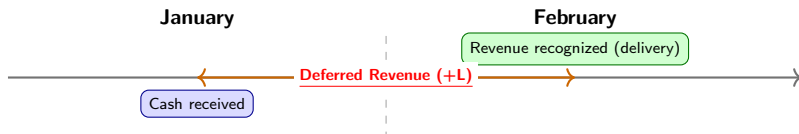
February:

Cash (+A)	10K
Acc.Rec (-A)	10K

Revenue Recognition - Case C

Prepare the **journal entries required at the end of each month:**

ABC corp. collected cash in January and delivered smartphones in February.



January:

Cash (+A)	10K
Def.Rev (+L)	10K

February:

Def.Rev (-L)	10K
Sales Revenues (+R, +SE)	10K

Revenues are recognized when products and services are delivered!

Revenues (R) - Formal

Increases in assets or decreases in liabilities that arise when a company provides goods or services.

- Case A. Revenues recognized with increases in assets (Cash).
- Case B. Revenues recognized with increases in assets (Accounts Receivable).
- Case C. Revenues recognized with decreases in liabilities (Deferred Revenues).

Matching Principle for Expense Recognition - Case A

Prepare the **journal entries required at the end of each month:**

ABC corp. received \$1K utility bill and paid it at the end of January.



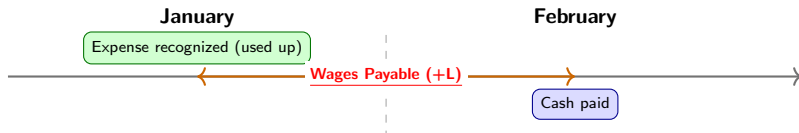
January:

Utility Expenses (+E)	1K
Cash (-A)	1K

Matching Principle for Expense Recognition - Case B

Prepare the **journal entries required at the end of each month**:

ABC corp.'s employees worked for the firm in January and received \$5K in February.



January:

Wages Expense (+E, -SE)	5K
Wages Payable (+L)	5K

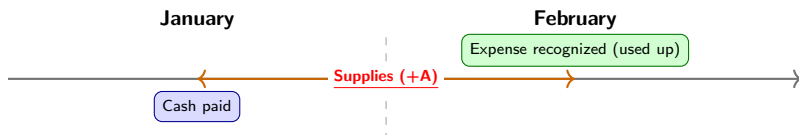
February:

Wages Payable (-L)	5K
Cash (-A)	5K

Matching Principle for Expense Recognition - Case C1

Prepare the **journal entries required at the end of each month:**

ABC corp. purchased supplies (\$2K) in January and used them in February.



January:

Supplies (+A)	2K
Cash (-A)	2K

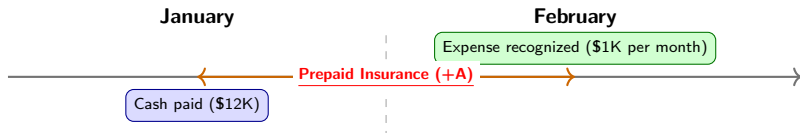
February:

Supplies Expense (+E)	2K
Supplies (-A)	2K

Matching Principle for Expense Recognition - Case C2

Prepare the **journal entries required at the end of each month**:

ABC corp. purchased a 12-months insurance plan (\$12K) at the beginning of month 1.



January:

Prepaid Insurance Expense (+A)	12K
Cash (-A)	12K

February:

Insurance Expense (+E)	1K
Prepaid Insurance Expense (-A)	1K

Expenses are recognized when “used up” / “incurred”

Expenses (E) - Formal

Decreases in assets or increases in liabilities relating to costs a company incurs when providing goods or services.

- Case A. Expenses recognized with decreases in assets (e.g., Cash).
- Case B. Expenses recognized with increases in liabilities (e.g., Accounts Payable; Wages Payable).
- Case C1 & C2. Expenses recognized with decreases in assets (e.g., Supplies; Prepaid Insurance).

- 1 Introduction to Income Statement and Accrual Accounting
 - Operating Activities and Cycle
 - Revenues, Expenses, and Net Income
 - Debit-Credit Framework - Expansion
 - Accrual-basis Accounting vs Cash-basis Accounting
- 2 Important Accounts
 - Accounts Receivable (Assets)
 - Accounts Payable (Liabilities)
 - Deferred Revenues (Unearned Revenues, Liabilities)
 - Prepaid Expenses (Assets)
- 3 Revenue and Expense Recognition - Cases
- 4 Illustration
 - Journal Entries
 - T-accounts
 - Unadjusted trial balance
- 5 Other topics
 - I/S - temporary & B/S - permanent
 - Net Profit Margin

- In Ch2, we have prepared journal entries for the following August transactions.
- In Ch3 (here), we prepare journal entries for the following September transactions.
- For B/S accounts: balances are carried over from August to September.
- For I/S accounts: balances are cleared at the end of August and initialized at the beginning of September.
- Why? revenues in August shouldn't be double-counted in September.

- 1 In September, Noodlecake sold \$3K of apps online. Cash collected in September.

Cash (+A)	3K
Revenues (+R, +SE)	3K

- 2 Noodlecake issued three \$100 gift cards at the beginning of September.

Cash (+A)	300
Def.Rev (+L)	300

- 3 Noodlecake sold \$9K of apps in the App Store and on Google Play. Cash will be collected in October.

Acc.Rec (+A)	9K
Revenues (+R, +SE)	9K

- 4 Noodlecake received checks totaling \$8,500 from Apple and Google, on account.

Cash (+A)	8,500
Acc.Rec (-A)	8,500

- 5 Noodlecake wrote checks to employees totaling \$7,800 for wages related to hours worked in September.

Wages Expense (+E, -SE)	7,800
Cash (-A)	7,800

- 6 On September 1, Noodlecake paid \$7,200 in advance for September, October, and November rent.

Prepaid Rent (+A)	7,200
Cash (-A)	7,200

- 7 Noodlecake displayed online ads in September and received a bill for \$500 to be paid in October.

Advertising Expenses (+E)	500
Acc.Pay (+L)	500

- 8 Noodlecake was notified by its bank that an automatic monthly payment of \$600 was transmitted to its utility company for electricity use in September.

Utility Expenses (+E)	600
Cash (-A)	600

T-accounts

Cash

Beg. 20,700	
3,000	7,800
300	7,200
8,500	600
End. 16,900	

Accounts Receivable

Beg. 0	
9,000	8,500
End. 500	

Supplies

Beg. 600	
End. 600	

Prepaid Rent

Beg. 0	
7,200	
End. 7,200	

Equipment

Beg. 9,600	
End. 9,600	

Software

Beg. 9,000	
End. 9,000	

Logo and Trademark

Beg. 300	
End. 300	

Accounts Payable

Beg. 10,200
500
End. 10,700

Deferred Revenues

Beg. 0
300
End. 300

Notes Payable

Beg. 20,000
0
End. 20,000

Common Stock

Beg. 10,000
End. 10,000

Retained Earnings

Beg. 0
See Ch4.

Sales Revenue

Beg. 0
3,000
9,000
End. 12,000

Wages Expense

Beg. 0
7,800
End. 7,800

Utilities Expense

Beg. 0
600
End. 600

Advertising Expense

Beg. 0
500
End. 500

Unadjusted trial balance

- Take all the debits and credits from the T-Accounts to the Trial Balance.

Account Title	Debit (\$)	Credit (\$)
Cash	16,900	
Accounts Receivable	500	
Supplies	600	
Prepaid Rent	7,200	
Equipment	9,600	
Logo & Trademarks	300	
Software	9,000	
Accounts Payable		10,700
Deferred Revenues		300
Notes Payable		20,000
Common Stock		10,000
Retained Earnings		0
Sales Revenue		12,000
Wages Expense	7,800	
Utilities Expense	600	
Advertising Expense	500	
Total	<u>53,000</u>	<u>53,000</u>

- 1 Introduction to Income Statement and Accrual Accounting
 - Operating Activities and Cycle
 - Revenues, Expenses, and Net Income
 - Debit-Credit Framework - Expansion
 - Accrual-basis Accounting vs Cash-basis Accounting
- 2 Important Accounts
 - Accounts Receivable (Assets)
 - Accounts Payable (Liabilities)
 - Deferred Revenues (Unearned Revenues, Liabilities)
 - Prepaid Expenses (Assets)
- 3 Revenue and Expense Recognition - Cases
- 4 Illustration
 - Journal Entries
 - T-accounts
 - Unadjusted trial balance
- 5 Other topics
 - I/S - temporary & B/S - permanent
 - Net Profit Margin

I/S - temporary & B/S - permanent

- I/S summarizes the financial effects of business activities from the current period.
- I/S accounts are summarized to Net Income.
- Net Income is taken to B/S through Retained Earnings.
- So, B/S includes the financial effects of business activities from all periods.
- Once I/S information is taken to B/S, I/S accounts are cleared (i.e., set to 0).
- I/S accounts then record R and E from the next period.

	<u>I/S</u>	<u>B/S</u>
Accounts	R, E	A, L, SHE
Represents the financial effects	just the current period	all periods
Beg. Bal.	0	from the previous period
Initialized?	yes	no
temporary or permanent?	temporary	permanent

Net Profit Ratio (or Net Profit Margin)

- A profitability ratio that measures how much net income is generated as a percentage of net sales.
- Calculated as: $\frac{\text{Net Income}}{\text{Net Revenues}}$

EX. Calculate Noodlecake's net profit margin.

- Net Income: \$3,100 (\$12K - \$7,800 - \$600 - \$500)
- Revenue: \$12K

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Revenues}} = \frac{3,100}{12,000} = 0.258 = 25.8\%$$